

Preclass Learning: Advanced Spatial Viz P1 (ICA 4)

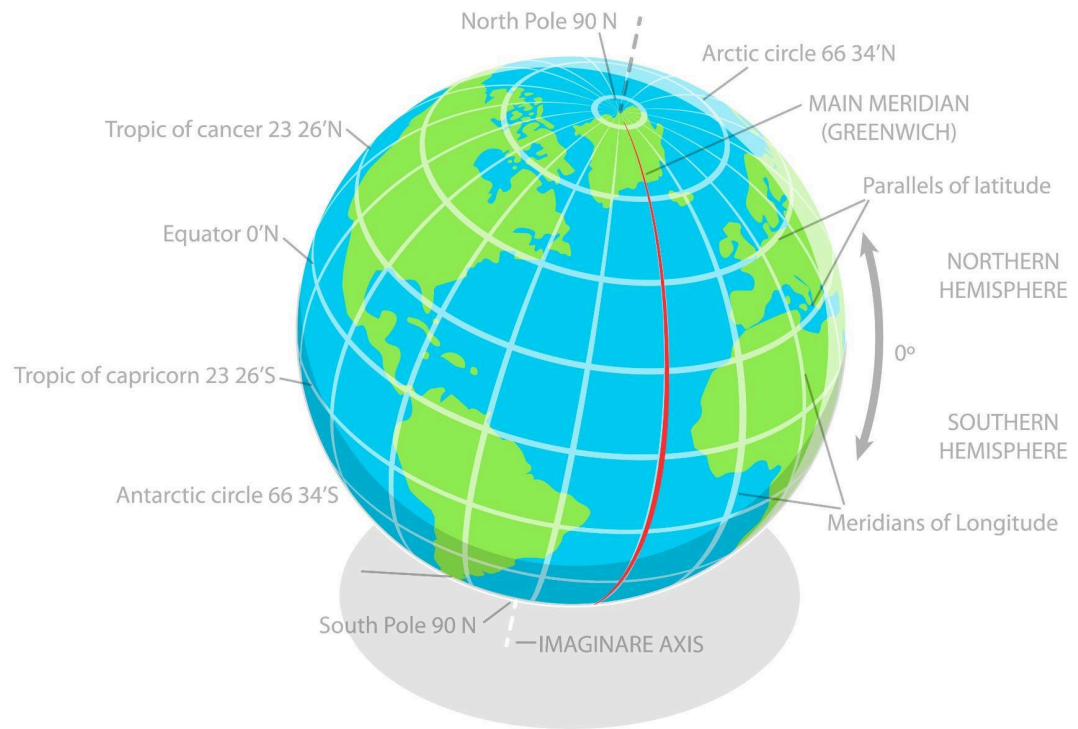
1. Geospatial Data Structures

- **Vector Data (Simple Features - SF):** Represents geographic features as distinct geometric shapes (points, lines, or polygons). Each feature has a row of attribute data associated with its geometry.
 - **Point:** A single location (X, Y coordinate).
 - **LineString:** A sequence of connected points (e.g., a road network).
 - **Polygon:** An enclosed area defined by an ordered list of coordinates (e.g., a city boundary).
- **Raster Data:** Represents geographic data as a grid of uniformly sized cells (pixels), where each cell holds a value (e.g., temperature, elevation).

2. Coordinate Reference Systems (CRS)

CRS defines how geographical locations are mathematically mapped to a 2D plane. It is the most critical element for accurate spatial analysis.

- **Geographic CRS:** Uses a 3D model (like a sphere) to define location using angular units (Latitude and Longitude, e.g., WGS 84).
- **Projected CRS:** Uses a 2D map projection defined by linear units (meters or feet). This is necessary to preserve specific properties like distance, area, or shape, but not all at once.
- **Transformation (Reprojection):** The process of converting spatial data from one CRS to another. **All spatial operations (e.g., joins, intersections) require the data layers to share the same CRS.**



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